

PROJECT PROPOSAL

Craft Base

Pakistan's Trusted Construction Company Directory

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Software Requirements Specification (SRS)

CraftBase — Construction Company Directory

1. Introduction

1.1 Purpose

This document specifies functional and non-functional requirements for the CraftBase system (CHS). CraftBase is Pakistan's dedicated online directory for construction companies, enabling homeowners, developers, and project managers to discover, evaluate, and hire verified construction contractors with confidence. The platform bridges a critical gap in Pakistan's construction industry by providing a centralized, transparent, and trust-driven marketplace.

1.2 Scope

The CraftBase system will:

- Allow clients (users) to:
 - Search and filter companies by city and trade specialization
 - Post project quote requests and receive competitive bids
 - Compare companies using trust badges and verified reviews
 - View project history and manage active quotes
 - Allow construction companies to:
 - Create and manage a verified company profile
 - Browse and bid on open project requests
 - View incoming bids and manage client relationships
 - Allow administrators to:
 - Verify company licenses and manage trust badges
 - Moderate community forum content
 - Manage user accounts and system data
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2. Overall System Description

2.1 System Architecture (High-Level)

- Client Layer (Multi-Page Web Application — HTML5/CSS3/JS)
- Shared Design System (style.css) — design tokens, responsive grid, reusable components

- Global JavaScript Utilities (main.js) — shared functions, mock data store, session simulation
- Page-Level Scripts — inline JS that consumes SAMPLE data and renders dynamic content
- Application Layer (REST API) — future backend integration (Node.js / Laravel)
- Database Layer (Relational DB: MySQL / PostgreSQL) — future persistent storage

2.2 User Roles

Role	Description
Client / User	Searches companies, posts projects, receives and compares bids, leaves reviews
Construction Company	Manages profile, browses open projects, submits competitive bids
Admin	Verifies licenses, assigns badges, moderates forum, manages all system data
Forum User	Participates in community discussions, asks questions, shares experiences

3. Functional Requirements

3.1 Client / User Module

3.1.1 User Registration & Authentication

FR-U1: User Registration

User shall register with:

- Full Name
- Email Address
- Password
- Account Type (Client or Company)
- System shall validate all inputs.
- Password must be hashed (e.g., bcrypt).

FR-U2: User Login

- User shall login using email and password.
- System shall generate session / token (JWT recommended).
- Navigation elements shall toggle between guest and authenticated views.

3.1.2 Company Search & Discovery

FR-U3: Search & Filter Companies

User shall:

- Enter keywords to search company names or specializations
- Filter results by city (Lahore, Karachi, Islamabad, etc.)
- Sort by rating, most hired, or most recent

System shall:

- Return real-time filtered results from the company directory
 - Display company cards with badges, ratings, and specialization tags
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3.1.3 Company Profile

FR-U4: View Company Profile

User shall view:

- Company name, description, specializations
 - Trust badges (Top Rated, Most Hired, Verified License)
 - Star ratings across 4 criteria with verified reviews
 - Project gallery and portfolio
 - Quote request button
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3.1.4 Quote & Project Workflow

FR-U5: Post a Project

Client shall:

- Select project type (Residential / Commercial / Renovation)
- Enter project description, budget range, city, and deadline
- Submit via a multi-step form

System shall:

- Validate all form inputs
- Store project as 'Open' and make it visible to registered companies

FR-U6: Compare & Hire

Client shall:

- View all received bids side-by-side
 - See bidding company's rating, badge context, and price
 - Select a winning bid and mark project as 'In Progress'
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3.1.5 User Dashboard

FR-U7: View Client Dashboard

User shall view:

- Active project requests and their current status
 - Received bids per project
 - Completed projects and review history
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3.1.6 Review System

FR-U8: Leave a Review

After project completion, client shall:

- Rate the company across 4 criteria (Quality, Timeline, Communication, Value)
- Write a verified review visible on the company profile

System shall:

- Calculate updated average rating
 - Re-evaluate company badge eligibility
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3.2 Construction Company Module

3.2.1 Company Registration (Self-Registration)

- Company shall self-register with: Company Name, Trade Specializations, Contact Info, License Number
- Admin shall review and verify license before badge assignment

3.2.2 Company Login

Company representative shall:

- Login using registered email and password
- Access company-specific dashboard

3.2.3 Bid Management

FR-C1: Submit Bid

Company shall:

- Browse all open project requests
- Submit a competitive quote with price and timeline

System shall:

- Attach bid to corresponding project request
- Notify the client of a new bid

FR-C2: Manage Profile

Company can:

- Update company description, gallery, and specializations
 - View badge status and criteria progress
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3.3 Admin Module

3.3.1 Badge Management

FR-A1: Assign / Revoke Badges

Admin shall:

- Review company credentials for Verified License badge
- System auto-assigns Top Rated badge (avg rating ≥ 4.5 with 10+ reviews)
- System auto-assigns Most Hired badge (selected in 20+ verified projects)

3.3.2 Forum Moderation

FR-A2: Moderate Forum

Admin shall:

- Pin and feature high-value threads
- Delete or flag inappropriate content
- Manage forum categories

3.3.3 User & Company Management

FR-A3: Manage Accounts

Admin shall:

- Activate or deactivate user and company accounts
- View all submitted projects and bids
- Permanently delete bookings or reviews

3.4 Community Forum Module

3.4.1 Forum Features

FR-F1: Browse & Create Threads

User shall:

- Browse threads by category (Advice, Reviews, Questions, Showcase, Market Trends)
- Search threads by keyword
- Create new threads with category, tags, and rich-text input
- Reply with threaded, nested comment support
- Upvote / downvote threads and replies for community curation

4. Project & Quote Status Lifecycle

Open (Posted by Client)



Bids Received (Companies Submit Quotes)



In Progress (Client Selects a Bid)



Completed (Project Finished)



Reviewed (Client Leaves Review)

Alternative transitions:

- Cancelled by Client
- Expired (No bids received before deadline)

5. Database Requirements (High-Level Schema)

5.1 Users Table

Field	Type
id	PK
name	VARCHAR
email	VARCHAR (Unique)
password_hash	TEXT
account_type	ENUM (client, company, admin)
created_at	TIMESTAMP

5.2 Companies Table

Field	Type
id	PK
user_id	FK
company_name	VARCHAR
city	VARCHAR
specializations	TEXT
license_no	VARCHAR

Field	Type
verified	BOOLEAN
avg_rating	DECIMAL

5.3 Projects Table

Field	Type
id	PK
client_id	FK
project_type	VARCHAR
description	TEXT
city	VARCHAR
budget_range	VARCHAR
deadline	DATE
status	ENUM (Open, In Progress, Completed, Cancelled)
created_at	TIMESTAMP

5.4 Bids Table

Field	Type
id	PK
project_id	FK
company_id	FK
price	DECIMAL
timeline	VARCHAR
status	ENUM (Pending, Accepted, Rejected)
submitted_at	TIMESTAMP

5.5 Reviews Table

Field	Type
id	PK

Field	Type
project_id	FK
client_id	FK
company_id	FK
quality_rating	INT
timeline_rating	INT
communication_rating	INT
value_rating	INT
review_text	TEXT
created_at	TIMESTAMP

5.6 Forum Threads Table

Field	Type
id	PK
user_id	FK
title	VARCHAR
category	ENUM
tags	TEXT
votes	INT
created_at	TIMESTAMP

6. Non-Functional Requirements

6.1 Performance

- System must support 500+ concurrent users.
- Page load time < 3 seconds on standard broadband.
- Real-time search filtering must respond in < 500ms.

6.2 Security

- Password hashing (bcrypt)
- JWT-based authentication for protected routes
- Role-based access control (RBAC): Client, Company, Admin

- Input validation and sanitization on all forms
- HTTPS mandatory

6.3 Availability

- 99% uptime target.
- Daily database backup.

6.4 Scalability

- Modular, component-based front-end architecture.
- API-first backend design for future mobile app extension.
- Deployable on cloud platforms (AWS / Azure / GCP).

6.5 Usability

- Fully responsive UI across mobile, tablet, and desktop screen sizes.
- Multi-step forms broken into logical steps to reduce friction.
- Max 3 steps for core workflows (project posting, registration).
- Dark Navy & Teal color palette for professional trust signaling.

7. Business Rules

1. A company cannot bid on the same project more than once.
2. Top Rated badge requires average rating ≥ 4.5 across all 4 criteria with 10+ verified reviews.
3. Most Hired badge requires selection in 20+ verified completed projects.
4. Verified License badge is assigned only after admin manual review.
5. Only Admin can permanently delete reviews or user accounts.
6. A client may only leave one review per completed project.
7. Projects expire and close automatically if no bids are received before the deadline.

8. Future Enhancements (Optional)

- Backend server integration (Node.js / Laravel / Django)
- Real database (PostgreSQL / MongoDB) with persistent storage
- Real-time notifications via WebSockets
- Payment gateway integration for premium company listings
- GPS-based contractor locator map
- Mobile app (iOS / Android) versions
- Admin panel for content moderation and license verification
- OAuth login (Google / Facebook)

9. Acceptance Criteria

System will be considered complete when:

- All 13+ HTML pages are implemented and functional
- All user roles (Client, Company, Admin) are tested
- Trust badge system displays correctly based on simulated performance metrics
- No critical UI or JavaScript errors
- Client can:
 - o Search and filter companies
 - o Post a project and view received bids
 - o Compare bids and manage dashboard
 - o Participate in the community forum

10. Suggested Tech Stack

Layer	Technology
Frontend	HTML5, CSS3, Vanilla JavaScript (ES6+)
CSS Architecture	Custom design system with CSS variables, Flexbox, CSS Grid
Backend (Future)	Node.js / Laravel / Django
Database (Future)	MySQL / PostgreSQL
Auth (Future)	JWT
Hosting	AWS / Azure / Netlify